

Solution Requirements

Date	10 JUNE 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system provides basic, session-based identity confirmation to partition working directories for student uploads and teacher configurations securely.	Medium

2	Authorization levels	Defines two discrete functional scopes: Students (access to record audio, view dashboard results, download PDFs) and Educators (access to manage master reference sheets)	High
3	External interfaces	Integrates seamlessly with local or remote OpenAI Whisper model configurations and Sentence-BERT endpoints for real-time natural language processing.	High
4	Transactions processing	Handles local disk staging operations safely, moving multi-format audio records securely through transcription and calculation queues without thread collisions.	High
5	Reporting	Automatically compiles analytical metrics into highly readable, download-ready PDF outputs containing graphical waveform signatures and structural summary tables.	High

6	Business rules, etc.	Computes grading outcomes based on clear conditional algorithms: cross-referencing semantic matching margins directly against strict delivery error thresholds (pause metrics and filler words).	High
7	Compliance to laws or regulations	Ensures safe localized handling of individual audio biometric captures and written records in compliance with fundamental digital data privacy protocols.	Medium
8	<i>Other</i>	Includes automated cache flushes within <code>/temp/</code> and <code>/uploads/</code> targets following the completion of active user operational sessions to save system overhead.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Audio processing runtime must support real-time user validation without causing frontend lag.	Whisper transcription and S-BERT matching execute in < 10 seconds for 1minute audio clips.

2	Scalability	System infrastructure structure must accommodate multithreaded analytical requests smoothly.	Supports concurrent processing operations for up to \$50\$ active instances without degradation.
3	Security & Data Privacy	Staged data elements must remain isolated and secure against unauthorized extraction protocols.	Local session storage uses protected paths; dynamic downloads expire immediately after session terminations.
4	Reliability & Availability	The system must be consistently available to users for submission processing and dashboard visualization.	Achieves a minimum operational runtime profile target of \$99.5\%\$ availability during system testing cycles.
5	Usability & Accessibility	Interface patterns must be intuitive and easily accessible across all responsive browser systems.	Streamlit web frontend passes standard fluid device responsiveness tests with high navigation clarity.
6	<i>Other</i>	Project architecture structure must favor clean isolation of processing layers to support codebase changes.	Code must adhere to strict modular decoupling standards across utility modules (speech_to_text, audio_utils).

